Laws
what must hold

$$\nabla \cdot \mathbf{D} = \rho_f, \quad \nabla \cdot \mathbf{B} = 0$$
$$\nabla \times \mathbf{E} = -\partial_t \mathbf{B}, \quad \nabla \times \mathbf{H} = \mathbf{J}_f + \partial_t \mathbf{D}$$

Ch. 3–6

Materials
how matter responds

$\mathbf{J}_c = \boldsymbol{\sigma}\mathbf{E}, \quad \mathbf{D} = \boldsymbol{\varepsilon}\mathbf{E}, \quad \mathbf{B} = \boldsymbol{\mu}\mathbf{H}$ or nonlinear B–H

may vary with position, direction, frequency, field

Ch. 3, 5, 13

Representation
choice of unknowns

$\mathbf{B} = \nabla \times \mathbf{A}, \quad \mathbf{E} = -\nabla\phi - \partial_t \mathbf{A}$

reference and gauge · finite-element spaces, mesh

Ch. 9, 16, 17

Observables
what is read out

$\mathbf{E}$ along a nerve · flux, terminal voltage · homogeneity

loss and SAR · force, torque · MRI phase

Ch. 18–22

Common ground: geometry · sources (coils, electrodes, magnets) · boundary and initial conditions